Week 14 – Sentiment Analysis Final Project

Learning Outcomes

By the end of this session, students will be able to:

1. To understand sentiment analysis
2. To perform text normalization, standardization, feature extraction for sentiment analysis through example
3. To identify and understand some lexicons for sentiment analysis
4. To use some of lexicons for sentiment analysis
5. Apply and use the different techniques for sentiment Analysis.

Introduction

In this project, your group will collect at least 500 Facebook/youtube/twitter/tiktok **short comments** from a specific domain (e.g., school-related issues, political drama, typhoon response, business reviews) and perform sentiment analysis using:

- Lexicon-based methods (VADER, TextBlob, AFINN)
- Traditional ML (TF-IDF + Logistic Regression or Naive Bayes)

You will compare the performance of these approaches and relate the findings to Affective Computing—how machines understand human emotions.

Each group must consist of 4 members. Regardless of roles, every member must participate in data collection (each member contributes a portion of the 500+ comments). Interpretation and analysis of results. Writing the final insights and reflections. Presenting the findings.

Ethical Data Collection Protocol

🛑 **Students must follow these rules when collecting Facebook comments:**

✔ **1. Collect only public comments**

Only gather data from:

- Public pages
- Public posts
- Public news outlets
- Publicly accessible community pages

Do NOT collect data from private groups or private individuals.

✔ **2. Do NOT collect names or personal identifiers**

Only store:

- The comment text

- The timestamp
- The reaction (optional)

Never store:

- Usernames
- Profile photos
- Links to personal accounts
- Personal information

✔ 3. Do NOT engage or respond to posts
Students should only observe and collect comments.
No interaction allowed.

✔ 4. If the topic is sensitive (politics, disaster, mental health)
Use **extra caution** and avoid storing:

- Personal stories
- Distressing or identifying details

✔ 5. Allowed Methods
You may use:

- **Manual copying** (copy comments to CSV)
- **FB comments export tools** (public posts only)
- **Web scraping tools** (only if permitted for public pages)

Automation must **not** bypass security or privacy restrictions.

Instructions:

Required Tasks

Task 1: Data Collection
Collect 500+ Facebook /Youtube/Twitter/Tiktok comments.
Deliverable:

- `raw_comments.csv` (text only, no names)

Task 2: Preprocessing
Perform:

- Cleaning (punctuation, URLs)
- Tokenization
- Stopword removal
- Lemmatization

Deliverable:

- `cleaned_comments.csv`

Task 3: Lexicon-Based Sentiment
Use at least **two** of:

- VADER

- TextBlob
- AFINN

Compute:

- Polarity scores
- Positive/Negative/Neutral labels

Deliverable:

- `lexicon_results.csv`
- Summary charts

Task 4: ML-Based Sentiment Analysis

Use:

- **TF-IDF vectorization**
- **Logistic Regression / Naive Bayes**

Evaluate using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

Deliverable:

- `ml_results.txt`
- `confusion_matrix.png`

Task 5: Comparative Analysis

Explain:

- Which method performed best?
- Why did lexicons differ from ML?
- What is the emotional distribution of your domain?
- How does this relate to affective computing?

Task 6: Final Presentation

1. Title Page

Group members, course, instructor, section, topic selected

2. Introduction

Explain your domain and why sentiment analysis matters

3. Dataset & Collection Process

- Source links
- Ethical steps taken
- Number of comments

4. Preprocessing Techniques

- Screenshots or summaries

5. Lexicon Analysis (VADER/TextBlob/AFINN)

- Sample outputs
- Sentiment distribution chart

6. ML-Based Analysis

- TF-IDF
- Logistic Regression/Naive Bayes results

- Confusion matrix

7. Comparison & Discussion

- Which model is best?
- Emotional patterns in your domain
- Challenges

8. Conclusion + Relation to Affective Computing

9. References

Department of Computer Science
College of Information Technology and Computing
CS 413 – CS Elective 2 (Affect Computing)
Laboratory Manual Submission Form

Sem and SY:	1st Sem, Sy 2025-2026	
Topic:		**Score:**
Week No(s):		
Date:		
Name:		**Course & Section:**

Answer Starts here: